Mahendra **Singh**

PhD Scholar · Atmospheric & Oceanic Sciences

Centre for Atmospheric and Oceanic Sciences (CAOS), Indian Institute of Science (IISc), Bangalore, India

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Research Profile

I am a fifth-year PhD research scholar at the Centre for Atmospheric and Oceanic Sciences (CAOS), Indian Institute of Science (IISc), Bangalore. My research focuses on Indian Summer Monsoon Low-Pressure Systems (LPSs), their climatology and representation in climate models, and their response to stratospheric aerosol geoengineering (SAG).
I use the CESM2.1.3 fully coupled Earth system model and the ERA5 reanalysis to investigate LPS characteristics and their responses to different climate scenarios, such as global warming and SAG.

Research Interests Indian Summer Monsoon; Low-Pressure Systems (LPSs); Monsoon Dynamics; Stratospheric Aerosol Geoengineering (SAG); Regional Climate Change

Research Experience

Centre for Atmospheric and Oceanic Sciences (CAOS), IISc *Bangalore, India*
PhD Research Scholar *2021 – Present*

- Investigating the climatology, dynamics, and representation of Indian Summer Monsoon Low-Pressure Systems (LPSs) in Earth system models.
- Evaluating LPSs genesis, tracks, frequency, intensity, duration, and rainfall contribution using CESM2.1.3 and ERA5 reanalysis.
- Diagnosing model biases through analysis of monsoon circulation, low-level jet structure, barotropic shear, dry-air intrusion, and atmospheric moisture transport.
- Investigating the response of monsoon LPSs to stratospheric sulfate aerosol geoengineering and its implications for regional rainfall and climate risk over South Asia.

Centre for Ocean, River, Atmosphere and Land Sciences (CORAL), IIT Kharagpur *Kharagpur, India*
M.Tech. Dissertation *2020 – 2021*

- Studied variability of upper-ocean characteristics in the North Indian Ocean region.
- Analysed oceanographic and atmospheric datasets using numerical and statistical methods.

Department of Physics, IIT Kanpur *Kanpur, India*
M.Sc. Dissertation *2016 – 2017*

- Investigated signatures of pseudo-scalar photon mixing on Cosmic Microwave Background radiation using WMAP and Planck data.

Publications

Journal Articles - Journal of Climate

- **Singh, M.**, Bala, G., & Seshadri, A. K. (2026). Accounting for the bias in the median track of Indian Summer monsoon low-pressure systems in an Earth system model. *Journal of Climate*. DOI: 10.1175/JCLI-D-25-0510.1.

Journal Articles - Accepted in Earth's Future

- **Singh, M.**, Bala, G., & Seshadri, A. K. (2026). An investigation of the characteristics of Indian Summer monsoon low-pressure systems under stratospheric aerosol geoengineering. *Earth's Future*. DOI: 10.1029/2026EF009141

Current Projects

Under Preparation

- **Singh, M.**, Bala, G., & Seshadri, A. K. ; Contrasting Moisture-Limited Monsoon Low-Pressure Systems under Single-Hemisphere Stratospheric Aerosol Geoengineering.

Conferences

106th AMS Annual Meeting — *Houston, Texas, USA*

Poster Presentation — *January 2026*

- Response of the Characteristics of Monsoon Low-Pressure Systems to Stratospheric Aerosol Geoengineering during the Indian Summer Monsoon Season.

EGU General Assembly 2025 — *Vienna, Austria*

Poster Presentation — *April 2025*

- Biases in model simulations of Indian Summer monsoon low-pressure systems.

Awards & Fellowships

Aug. 2022 – Present	**Prime Minister's Research Fellowship (PMRF),** Government of India	*India*
2022	**Grantham Fellowship,** Awarded for Academic Excellence	*IISc, Bangalore*
2017	**CSIR National Eligibility Test – JRF,** Physics	*India*
2019	**Graduate Aptitude Test in Engineering (GATE),** Physics – AIR 142	*India*
2018	**Joint Entrance Screening Test (JEST),** AIR 38	*India*
2012	**INSPIRE Scholarship,** Department of Science and Technology, Government of India	*India*

Education

Indian Institute of Science (IISc) — *Bangalore, India*

PhD in Atmospheric and Oceanic Sciences — *2021 – Present*

- Research on Indian Summer Monsoon LPSs, and Stratospheric Aerosol Geoengineering (SAG).
- CGPA: 8.90/10; Centre for Atmospheric and Oceanic Sciences (CAOS).

Indian Institute of Technology Kharagpur — *Kharagpur, India*

M.Tech. in Earth System Science and Technology — *2019 – 2021*

- CGPA: 8.91/10; Centre for Ocean, River, Atmosphere and Land Sciences (CORAL).

Indian Institute of Technology Kanpur — *Kanpur, India*

M.Sc. in Physics — *2015 – 2017*

- CPI: 6.7/10.

Chhatrapati Shahu Ji Maharaj University — *Kanpur, India*

B.Sc. in Physics, Mathematics and Chemistry — *2012 – 2015*

Technical Skills

Programming	Python, MATLAB, FORTRAN, NCL
Climate Models	CESM2.1.3, CAM6, Coupled Earth system modelling
Data Analysis	ERA5, CMIP datasets, NetCDF, xarray, NumPy, SciPy
Visualization	Matplotlib, Cartopy, NCL, Scientific visualization
Research Methods	Atmospheric dynamics, Moisture budget analysis, Synoptic-system tracking, Climate-model evaluation, Statistical analysis